

Aphasia Severity Prediction from Multi-Task Spontaneous Speech Using Interpretable Linguistic Features

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Abstract

This study investigates whether simple, interpretable linguistic features extracted from spontaneous speech can predict aphasia severity across multiple discourse tasks. Using AphasiaBank transcripts from multiple corpora, we analyze three picture description tasks (Cat, Umbrella, Window) and extract Mean Length of Utterance (MLU), Type-Token Ratio (TTR), and function-word ratio. Aphasia severity is operationalized using the Aphasia Quotient (AQ), binarized as $AQ < 75$ versus $AQ \geq 75$. To prevent participant-level leakage across tasks, we employ a grouped train-test split. A Logistic Regression model achieves an accuracy of 0.74 on the held-out test set, comparable to prior machine learning approaches using more complex feature sets. Standardized coefficients show that MLU is the strongest predictor, while TTR and function-word ratio contribute more weakly. These results indicate that simple, interpretable features can capture clinically relevant variation in speech samples from people with aphasia (PWA). At the same time, observed variability across tasks highlights the importance of accounting for discourse type in aphasia severity modeling.

1 Introduction

Aphasia severity is a central clinical variable that informs diagnosis, treatment planning, and prognosis. It is typically quantified using standardized assessments such as the Western Aphasia Battery, which yields summary measures like the Aphasia Quotient (AQ). While such instruments provide reliable and widely used benchmarks, they are time-intensive and constrained to structured testing contexts. As a result, they do not directly capture how individuals use language in spontaneous, ecologically valid settings. Language sample analysis (LSA) has been proposed as a more ecologically valid approach for assessing language disorders, as it captures real-world language use be-

yond structured testing (MacWhinney and Fromm, 2022). This gap has motivated increasing interest in computational approaches that estimate aphasia severity from naturalistic speech.

Recent work in clinical natural language processing (NLP) has explored automated analysis of speech samples of PWA using a range of linguistic and acoustic features. These approaches include both complex feature representations and models leveraging machine learning techniques for severity prediction (Day et al., 2021; Kristinsson et al., 2021). However, such methods often require substantial amounts of data and may be difficult to interpret in clinically meaningful terms. In contrast, simpler linguistic features remain attractive in small-sample settings typical of aphasia research, as they offer transparency, theoretical grounding, and ease of replication. At the same time, existing studies have largely focused on single elicitation tasks and individual corpora, limiting our understanding of how well these features generalize across different contexts.

AphasiaBank provides a large collection of transcribed speech samples elicited through standardized discourse tasks, enabling systematic analysis of spontaneous speech across participants and studies (MacWhinney et al., 2011). These tasks include picture descriptions such as Cat, Umbrella, and Window, which are widely used in aphasia research. However, different discourse elicitation tasks impose distinct cognitive and linguistic demands, leading to systematic variation in linguistic features such as lexical diversity, speech rate, and syntactic structure (Deng et al., 2024). This raises the possibility that feature-severity relationships may be task-dependent. This introduces a key methodological question: whether simple linguistic features that perform well in one task remain robust across tasks, and whether combining multiple tasks enhances or degrades predictive performance due to increased variability.

The present study addresses these issues by evaluating the extent to which a small set of interpretable linguistic features can predict aphasia severity from spontaneous speech across multiple AphasiaBank corpora. We focus on Mean Length of Utterance (MLU), Type-Token Ratio (TTR), and function-word ratio, which are standard measures derived from language sample analysis and implemented in tools such as TalkBank (MacWhinney and Fromm, 2022). Prior work has shown that measures such as TTR can be sensitive to sample characteristics and task variation (Fergadiotis et al., 2013), further motivating careful evaluation across tasks. Specifically, we ask: (1) to what extent can simple linguistic features derived from spontaneous speech predict aphasia severity in picture description tasks? and (2) how does predictive performance change when moving from a single-task setting to a multi-task setting?

2 Data

The data were drawn from multiple corpora within AphasiaBank, a large, standardized database of discourse samples for individuals with and without aphasia (MacWhinney et al., 2011; MacWhinney and Fromm, 2022). AphasiaBank provides systematically collected language samples across a range of elicitation tasks, enabling cross-study comparability and large-scale analysis of spontaneous speech. We adopted an inclusion-first strategy: all available corpora were audited, and file-level inclusion criteria were applied uniformly. Files were retained if they contained a valid participant AQ score, at least one analyzable target task, and participant utterances suitable for feature extraction. The final analysis used participants with valid observations for all three target tasks: Cat, Umbrella, and Window, which are standard picture description tasks widely used in discourse analysis within AphasiaBank. All corpora analyzed in this study are part of the AphasiaBank repository and follow a shared discourse elicitation protocol and transcription standard, ensuring comparability across sites (MacWhinney et al., 2011; MacWhinney and Fromm, 2022).

The final dataset contained 396 participants and 1,188 participant-task observations. Each participant contributed one observation for each of the three target tasks. Severity labels were approximately balanced, with 196 $AQ < 75$ observations and 200 $AQ \geq 75$ observations per task.

3 Methods

The unit of analysis was the participant-task pair. Aphasia severity was operationalized using the Aphasia Quotient (AQ), extracted from the participant @ID line of each CHAT file. AQ was binarized following standard clinical thresholds, where $AQ < 75$ was coded as 0 and $AQ \geq 75$ was coded as 1.

Task segments were identified using CHAT G-coded task markers within the TalkBank transcription format (MacWhinney and Fromm, 2022). Only participant utterances were analyzed. Tokens were lowercased, and only alphabetic tokens were retained. Utterances containing no alphabetic tokens were excluded prior to feature computation to avoid distortion from punctuation-only or non-linguistic content. No lemmatization or additional normalization was applied in order to preserve surface-level variation in spontaneous speech.

For each participant-task pair, we represent the language sample as a sequence of utterances:

$$U = \{u_1, u_2, \dots, u_M\}$$

where M is the total number of utterances. Each utterance u_i consists of a sequence of tokens:

$$u_i = \{w_{i1}, w_{i2}, \dots, w_{iL_i}\}$$

where L_i is the number of tokens in utterance i . The total number of tokens in the sample is:

$$N_{\text{tokens}} = \sum_{i=1}^M L_i$$

Three linguistic features were extracted.

Mean Length of Utterance (MLU) Mean Length of Utterance captures average utterance length and is defined as:

$$MLU = \frac{1}{M} \sum_{i=1}^M L_i = \frac{N_{\text{tokens}}}{M}$$

MLU is a standard measure of syntactic and structural complexity in language sample analysis (MacWhinney and Fromm, 2022).

Type-Token Ratio (TTR) Let V denote the set of unique word types in the sample:

$$V = \{w : w \in U\}$$

with cardinality $|V| = N_{\text{types}}$. Type-Token Ratio is defined as:

$$TTR = \frac{N_{\text{types}}}{N_{\text{tokens}}}$$

Corpus	Participants	Observations	Cat	Umbrella	Window
Kurland	61	183	61	61	61
SCALE	54	162	54	54	54
UNH	34	102	34	34	34
Adler	25	75	25	25	25
Williamson	25	75	25	25	25
Kansas	23	69	23	23	23
Tucson	22	66	22	22	22
Richardson	21	63	21	21	21
Whiteside	19	57	19	19	19
Fridriksson	17	51	17	17	17
Thompson	16	48	16	16	16
MCU	15	45	15	15	15
UMD	14	42	14	14	14
ACWT	11	33	11	11	11
Baycrest	10	30	10	10	10
BU	10	30	10	10	10
Wozniak	7	21	7	7	7
Wright	7	21	7	7	7
Kempler	3	9	3	3	3
Garrett	2	6	2	2	2
Total	396	1188	396	396	396

Table 1: Corpus-level composition of the final dataset after AQ filtering, task filtering, and restriction to participants with all three tasks.

Task	AQ < 75	AQ ≥ 75	Total
Cat	196	200	396
Umbrella	196	200	396
Window	196	200	396

Table 2: Task-level severity distribution in the final dataset.

TTR measures lexical diversity but is known to be sensitive to sample length and discourse characteristics (Fergadiotis et al., 2013).

Function-word ratio Let $\mathcal{F}$ denote the set of function words (e.g., determiners, conjunctions, prepositions, auxiliary verbs, pronouns, and wh-words). The number of function word tokens is:

$$N_{\text{function}} = \sum_{i=1}^M \sum_{j=1}^{L_i} I(w_{ij} \in \mathcal{F})$$

where $I(\cdot)$ is the indicator function. The function-word ratio is defined as:

$$\text{Function-word ratio} = \frac{N_{\text{function}}}{N_{\text{tokens}}}$$

This feature captures the relative use of grammatical versus content words, which has been shown to vary with aphasia severity.

Modeling We used Logistic Regression as the primary model due to its interpretability and suit-

ability for low-dimensional feature spaces, as commonly employed in clinical NLP applications (Day et al., 2021). Let $\mathbf{x}_i \in \mathbb{R}^3$ denote the feature vector for observation i :

$$\mathbf{x}_i = [\text{MLU}_i, \text{TTR}_i, \text{FWR}_i]$$

The model estimates:

$$P(y_i = 1 \mid \mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i + b)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic function, $\mathbf{w}$ are feature weights, and b is the bias term.

Features were standardized using z-score normalization:

$$x' = \frac{x - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation computed on the training set.

To prevent leakage across tasks from the same participant, we used a grouped train-test split based on participant identifiers, ensuring that all observations from a given participant were assigned exclusively to either the training or test set. The data were split using an 80/20 ratio with a fixed random seed for reproducibility.

4 Results

Table 3 reports feature means by task and severity. MLU shows a consistent separation between severity groups across all tasks. Participants with AQ

Task	Severity	MLU	TTR	Func.word ratio
Cat	AQ < 75	4.077	0.601	0.353
Cat	AQ ≥ 75	6.915	0.564	0.440
Umbrella	AQ < 75	3.429	0.586	0.313
Umbrella	AQ ≥ 75	5.354	0.575	0.408
Window	AQ < 75	3.590	0.623	0.334
Window	AQ ≥ 75	6.153	0.625	0.431

Table 3: Mean feature values by task and severity group.

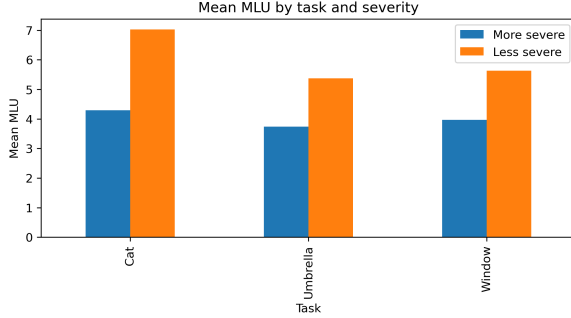


Figure 1: Mean MLU by task and severity group.

≥ 75 produced longer utterances than participants with AQ < 75 in Cat, Umbrella, and Window, consistent with prior findings that utterance length is sensitive to aphasia severity (Deng et al., 2024).

In contrast, TTR does not show a consistent pattern across severity groups. While slightly lower TTR values are observed in the AQ ≥ 75 group for some tasks, the differences are small and not uniform, which aligns with prior observations that TTR is sensitive to sample length and task characteristics (Fergadiotis et al., 2013). Function-word ratio shows a more consistent increase in the less severe group across all tasks.

The Logistic Regression model achieved an accuracy of 0.74 on the held-out test set. Performance was moderately balanced across classes, with a macro-averaged F1 score of 0.73. This level of performance is comparable to prior work using machine learning models on AphasiaBank transcripts, which has reported accuracies in the range of approximately 0.70–0.75 (Day et al., 2021).

Standardized coefficients are shown in Table 5. MLU had the largest coefficient, indicating that it was the strongest predictor of the less severe class. Function-word ratio and TTR also had positive coefficients, but their effects were smaller. This suggests that structural properties of language production contribute more strongly to severity classification than lexical diversity in this feature space.

Label	Precision	Recall	F1	Support
AQ < 75	0.699	0.657	0.677	99
AQ ≥ 75	0.769	0.801	0.785	141
Macro avg.	0.734	0.729	0.731	240
Weighted avg.	0.740	0.742	0.740	240

Table 4: Classification performance of the standardized Logistic Regression model. Overall accuracy was 0.742.

Feature	Coefficient
MLU	1.258
Function-word ratio	0.377
TTR	0.231

Table 5: Standardized Logistic Regression coefficients. Positive coefficients indicate higher probability of AQ ≥ 75.

5 Discussion

The results indicate that simple linguistic features extracted from spontaneous speech can predict aphasia severity above chance in a multi-corpus, multi-task setting. MLU emerged as the strongest predictor, supporting the hypothesis that utterance-level structural productivity is a robust marker of aphasia severity. This pattern was consistent across the three picture description tasks and aligns with prior findings showing that measures of utterance length reliably differentiate between severity levels in discourse production (Deng et al., 2024).

TTR showed weaker and less stable behavior. In some tasks, the more severe group showed comparable or even higher TTR values, likely because shorter samples can inflate lexical diversity estimates. This interpretation is consistent with previous work demonstrating that TTR is highly sensitive to sample length and may yield misleading estimates in short or constrained discourse samples (Fergadiotis et al., 2013). Function-word ratio showed a more consistent positive association with the less severe class, but its contribution was smaller than that of MLU. These findings suggest that simple lexical ratios should be interpreted cautiously, especially in short or sparse speech samples.

The multi-task design provides a stricter test of generalizability than a single-task analysis. Prior work has shown that discourse task type significantly influences linguistic measures, including lexical diversity, syntactic structure, and word class distribution (Deng et al., 2024). Although the model achieved moderate accuracy, task variabil-

ity remains an important source of noise. This is consistent with existing findings that different elicitation tasks impose distinct cognitive and linguistic demands, leading to variability in observed feature distributions.

Compared to prior machine learning approaches that rely on higher-dimensional or less interpretable feature sets (Day et al., 2021; Kristinsson et al., 2021), the present results demonstrate that a small set of interpretable linguistic features can achieve comparable predictive performance. This suggests that relatively simple representations of language production may capture clinically relevant variation in PWA’s speech without requiring complex modeling pipelines. The findings therefore support the use of interpretable features for severity prediction, while also showing that task structure and corpus heterogeneity must be explicitly controlled. These results are not intended to replace standardized clinical assessments, but rather to complement them by providing scalable and interpretable estimates of severity from spontaneous speech.

Limitations

This study has several limitations. First, the analysis uses a binary AQ threshold, which simplifies a continuous clinical measure and may obscure more fine-grained variation in aphasia severity. Prior work has shown that severity exists along a continuum and can be modeled using regression-based approaches (Day et al., 2021; Kristinsson et al., 2021). Second, the feature set is intentionally limited and does not include syntactic parses, semantic content measures, acoustic features, or discourse-level variables. While this was motivated by a focus on interpretability, it likely restricts the upper bound of predictive performance.

Third, the study uses a single grouped train–test split rather than cross-validation. Although grouping by participant prevents leakage across tasks, future work should evaluate robustness across multiple splits or cross-validation schemes. Fourth, although the final dataset is larger than many aphasia NLP datasets, corpus contributions are uneven, which may introduce bias related to corpus-specific elicitation or transcription practices.

Finally, the analysis depends on the consistency of CHAT task annotations across corpora. Although AphasiaBank provides standardized protocols and transcription guidelines (MacWhinney et al., 2011; MacWhinney and Fromm, 2022), vari-

ation in annotation practices and task implementation may still introduce noise into feature extraction and downstream modeling.

6 Conclusion

This study evaluated whether simple linguistic features from spontaneous speech can predict aphasia severity across multiple AphasiaBank corpora and picture description tasks. Logistic Regression achieved moderate classification performance, and standardized coefficients showed that MLU was the strongest predictor. This result is consistent with prior work demonstrating that structural measures of language production are sensitive to aphasia severity in discourse contexts (Deng et al., 2024).

More broadly, these findings suggest that interpretable linguistic features derived from language sample analysis can capture clinically relevant variation in PWA’s speech without requiring complex modeling pipelines (MacWhinney and Fromm, 2022). At the same time, the results highlight the importance of task variability and cross-corpus comparability, as differences in elicitation methods can substantially affect linguistic measures. Future work should extend this approach to richer feature sets and more robust evaluation frameworks, while maintaining interpretability and methodological rigor.

Code and Data Availability

Code and processed data for this study are available at: https://github.com/aekeser/NLP_Project_Aphasia. Due to licensing restrictions, the original AphasiaBank data are not redistributed. Access to the raw data can be obtained through TalkBank under its data use agreement.

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A Reproducibility Details

The final dataset contains 396 participants and 1,188 observations. The train–test split was grouped by participant identifier and saved to ensure exact reproducibility.